Use: Amylase and pH: Continuous-Sampling Investigation

Today I can… ☐ Put the steps in order: get ready, start timing, then take samples. Give a reason for one choice about evidence. ☐ Find the first iodine endpoint seen. Record its time. State the limit caused by the time gap between samples. ☐ Use the raw results to compare. Match the conditions. Give a comparison that the observations support.

Feedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R ☐ [//] ☐ [?] _____

Arrival · Supported

1. Enzyme and substrate

Link to today: Amylase is the enzyme; starch is its substrate. Which one speeds up the breakdown reaction?

An enzyme is a biological catalyst: it speeds up a reaction without being used up.

2. Read the colour

Iodine becomes blue-black with starch. A sampled drop turns blue-black: is starch detected?

Blue-black = starch detected. Orange-brown = starch not detected.

3. Recall binding

Which substrate fits amylase’s active site: starch or an unrelated substrate?

Recall the enzyme and substrate model from the previous lesson.

4. Prepare the evidence

Which comes first: prepare the iodine wells / mix and start timing?

The reaction continues after mixing, so the sampling equipment must already be ready.

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Exit Ticket · Supported

Evidence

These are example results at pH 7: 30 s blue-black; 60 s blue-black; 90 s orange-brown. Blue-black means starch was detected. Orange-brown means no starch was detected in that sample. The planned interval is 30 s. The amounts at the start match. The temperature and the endpoint rule match too.

1. Choose the first observed endpoint time: 60 s or 90 s. What does its colour tell you?

Use the colour key. Endpoint = first sample with starch not detected.

2. A friend writes 75 s exactly. Use the sampled times to help them repair it.

Choose an observed time. Say “We did not observe …”.

Under the heading Exit ticket: answer the short questions in your book.